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453 CIS-4 NETWORK SECURITY

Project

1. Visit a login page in any trusted website and use Wireshark to : (2 Marks)

Show the data associated with each OSI layer in details.

2- do the same task but with http page (not encrypted) and show the differences between layers.

Handwritten notes on the Wireshark interface:

- Red box around 'http' in the packet list.
- Red arrow pointing to the packet details pane with the text: "Not Encrypted → TLS/SSL Protocol use for Encryption".
- Red arrow pointing to the packet details pane with the text: "Data Link".
- Blue arrow pointing to the packet details pane with the text: "Application Layer".
- Blue arrow pointing to the packet details pane with the text: "Network Layer".
- Blue arrow pointing to the packet details pane with the text: "Transport Layer".

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
51	61.064254601	10.0.2.15	216.58.211.206	HTTP	375	GET / HTTP/1.1
53	61.168782756	216.58.211.206	10.0.2.15	HTTP	786	HTTP/1.1 301 Moved Permanently (text/html)
62	61.286877503	10.0.2.15	172.217.18.36	HTTP	379	GET / HTTP/1.1
64	61.425554593	172.217.18.36	10.0.2.15	HTTP	1041	HTTP/1.1 302 Found (text/html)
88	61.808901881	10.0.2.15	216.58.211.195	OCSP	470	Request
90	62.018816479	216.58.211.195	10.0.2.15	OCSP	755	Response
180	62.426510718	10.0.2.15	216.58.211.195	OCSP	471	Request

Packet Details:

- Frame 51: 375 bytes on wire (3000 bits), 375 bytes captured (3000 bits) on interface eth0, id 0
- Ethernet II, Src: PcsCompu_22:46:4f (08:00:27:22:46:4f), Dst: RealtekU_12:35:02 (52:54:00:12:35:02)
- Internet Protocol Version 4, Src: 10.0.2.15, Dst: 216.58.211.206
- Transmission Control Protocol, Src Port: 45868, Dst Port: 80, Seq: 1, Ack: 1, Len: 321
- Hypertext Transfer Protocol
 - GET / HTTP/1.1\r\n
 - Host: google.com\r\n
 - User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0\r\n
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n
 - Accept-Language: en-US,en;q=0.5\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Connection: keep-alive\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - \r\n
 - [Full request URI: http://google.com/]
 - [HTTP request 1/1]
 - [Response in frame: 53]

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 - Host: google.com\r\n
 - User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:91.0) Gecko/20100101 Firefox/91.0\r\n
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n
 - Accept-Language: en-US,en;q=0.5\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Connection: keep-alive\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - \r\n
 - [Full request URI: http://google.com/]
 - [HTTP request 1/1]
 - [Response in frame: 53]

We have 7 Layers of OSI Model

#1. The Physical Layer

bitstream is physically transmitted over a physical medium

#2. The Data Link Layer

transmitted into frames for transmission at the physical

#3. The Network Layer

operate on packets and are responsible for routing traffic to its destination based on IP addresses.

#4. The Transport Layer

“host” layers with the rest referred to as “media” layers.

is the “segment” or “datagram”. This layer manages the transmission of data between nodes

#5. The Session Layer

manages sessions between nodes and acts on the “data”

#6. The Presentation Layer

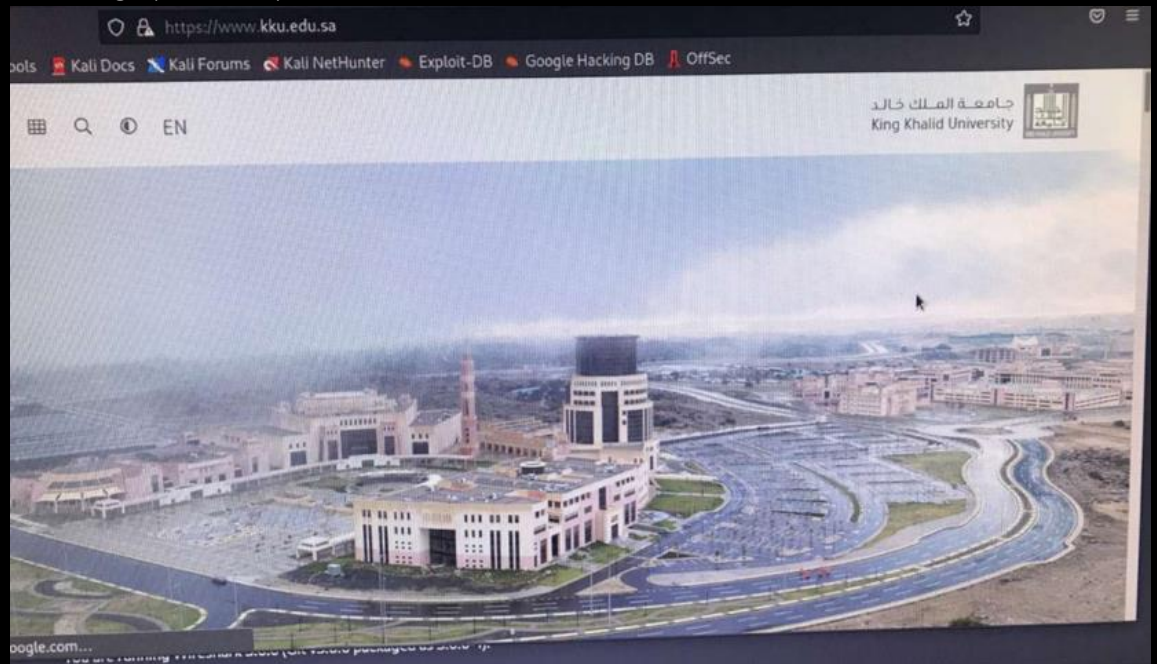
translating data from network data to the formats expected by an application.

#7. The Application Layer

includes protocols designed for end-users.

TLS/SSL is the most common encryption protocol, which is used every day on the Internet. TLS/SSL stands for Transport Layer Security/ Secure Sockets Layer, which is an encryption protocol that ensures communications between a client and server are kept secure.

2. Establish a handshake session with any trusted website and show the following : (3 Marks)



1 . What is the IP address of the website >>use nslookup in command terminal .

```
Command Prompt
Microsoft Windows [Version 10.0.19045.2364]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>nslookup www.kku.edu.sa
Server:  bind-dns02.kku.edu.sa
Address:  10.6.16.142

Name:     www.kku.edu.sa
Address:  10.4.8.68

C:\Users\Student>
```

Then from Wireshark, inspect the following:

2. Client Hello message and show cipher suite supported by your browser .

Time	Source	Destination	Protocol	Length	Info
34 6.634735795	10.0.2.15	216.58.205.206	TLSv1.3	716	Client Hello
35 6.635283527	10.0.2.15	216.58.205.206	TLSv1.3	66	Change Cipher Spec
36 6.635373985	10.0.2.15	216.58.205.206	TLSv1.3	224	Application Data
37 6.635434038	216.58.205.206	10.0.2.15	TCP	60	443 → 47872 [ACK] Seq=1 A
38 6.636269549	216.58.205.206	10.0.2.15	TCP	60	443 → 47872 [ACK] Seq=1 A
39 6.636269904	216.58.205.206	10.0.2.15	TCP	60	443 → 47872 [ACK] Seq=1 A
40 6.655295005	216.58.205.206	10.0.2.15	QUIC	137	Protected Payload (KPO),

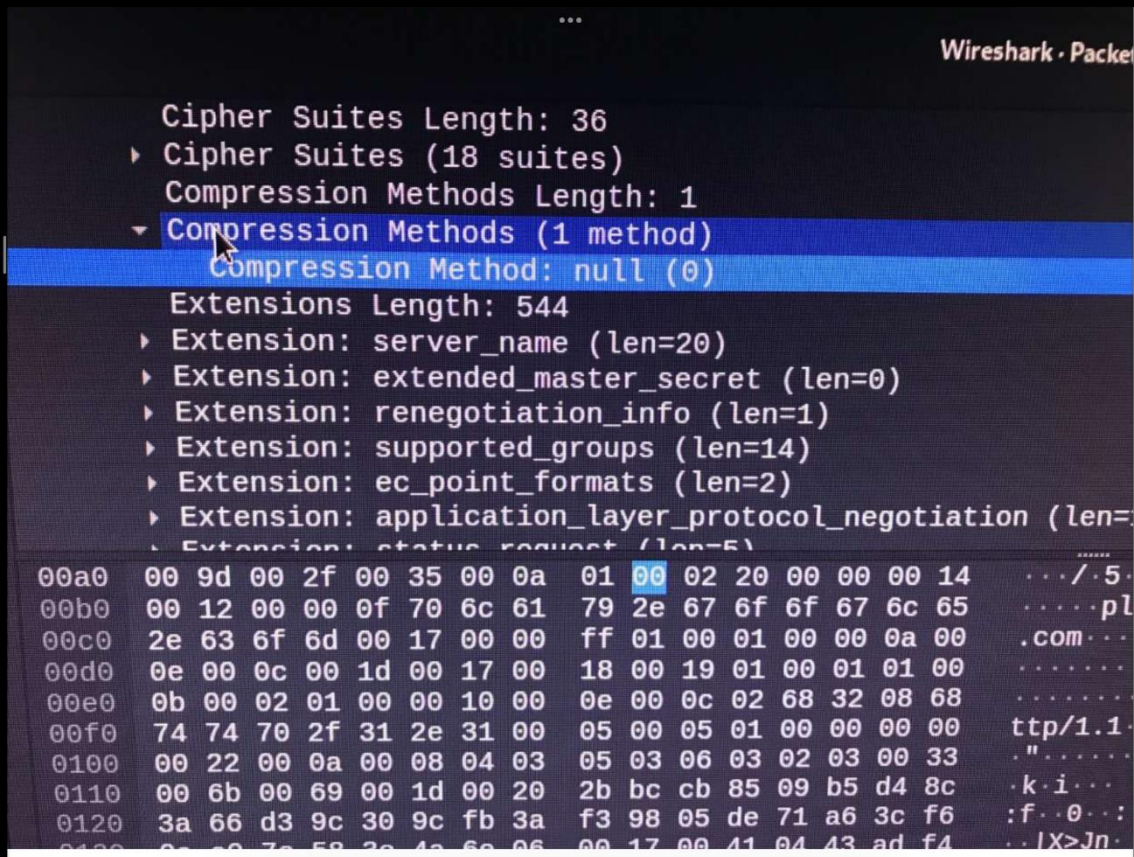
Frame 34: 716 bytes on wire (5728 bits), 716 bytes captured (5728 bits) on interface eth0, id 0
Ethernet II, Src: PcsCompu_22:46:4f (08:00:27:22:46:4f), Dst: RealtekU_12:35:02 (52:54:00:12:35:02)
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 216.58.205.206
Transmission Control Protocol, Src Port: 47872, Dst Port: 443, Seq: 1, Ack: 1, Len: 662
Transport Layer Security

Cipher Suites Length: 36					
- Cipher Suites (18 suites)					
Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)					
Cipher Suite: TLS_CHACHA20_POLY1305_SHA256 (0x1303)					
Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)					
Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (0xc02b)					
Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)					
Cipher Suite: TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca9)					
Cipher Suite: TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca8)					
Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 (0xc02c)					
Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (0xc030)					
Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA (0xc00a)					
Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA (0xc009)					
0080	f2 d4 00 24	13 01 13 03	13 02 c0 2b	c0 2f cc a9	...\$. +/..
0090	cc a8 c0 2c	c0 30 c0 0a	c0 09 c0 13	c0 14 00 9c	..., -0..
00a0	00 9d 00 2f	00 35 00 0a	01 00 02 20	00 00 00 14	.../.5..
00b0	00 12 00 00	0f 70 6c 61	79 2e 67 6f	6f 67 6c 65	...pla y.google
00c0	2e 63 6f 6d	00 17 00 00	ff 01 00 01	00 00 0a 00	.com.....
00d0	0e 00 0c 00	1d 00 17 00	18 00 19 01	00 01 01 00	
00e0	0b 00 02 01	00 00 10 00	0e 00 0c 02	68 32 08 68	h2.h
00f0	74 74 70 2f	31 2e 31 00	05 00 05 01	00 00 00 00	ttp/1.1.
0100	00 22 00 0a	00 08 04 03	05 03 06 03	02 03 00 33	3
0110	00 6b 00 69	00 1d 00 20	2b bc cb 85	09 b5 d4 8c	.k.i... +.....
0120	3a 66 d3 9c	30 9c fb 3a	f3 98 05 de	71 a6 3c f6	:f..θ...: ...q.<
0130	0a e0 7c 58	3e 4a 6e 06	00 17 00 41	04 43 ad f4	.. X>Jn...A.C..
0140	5c d2 14 96	4c 8b ea 4b	ca 9d 11 48	ab c9 fa 95	\...L..K ...H...
0150	55 fa b6 34	2a 4b 5a f7	80 a7 e8 86	75 5a 89 84	U...4*KZ...uZ...
0160	4d 1c 75 5c	7b ba e8 d2	4c 93 be bc	2e 4d b2 55	M.u\{... L...M.U

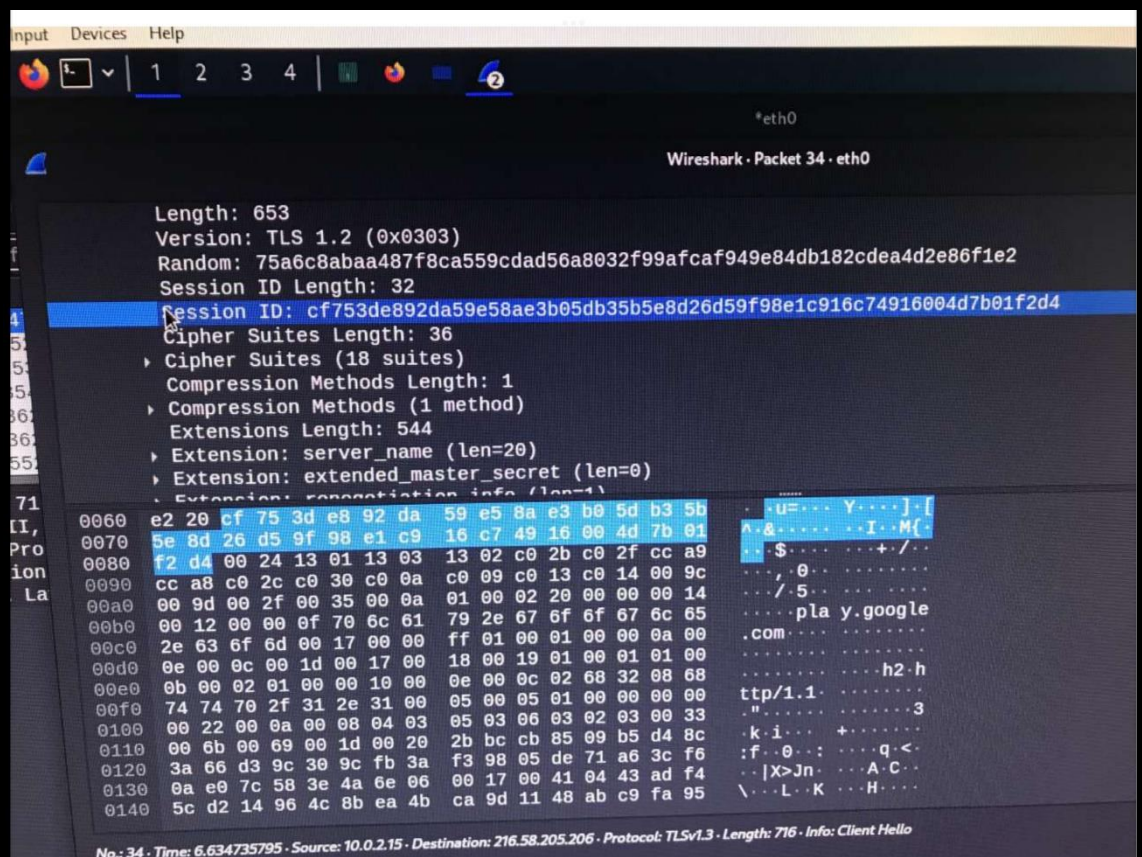
No.: 34 - Time: 6.634735795 - Source: 10.0.2.15 - Destination: 216.58.205.206 - Protocol: TLSv1.3 - Length: 716 - Info: Client Hello

3. What is the compression method?

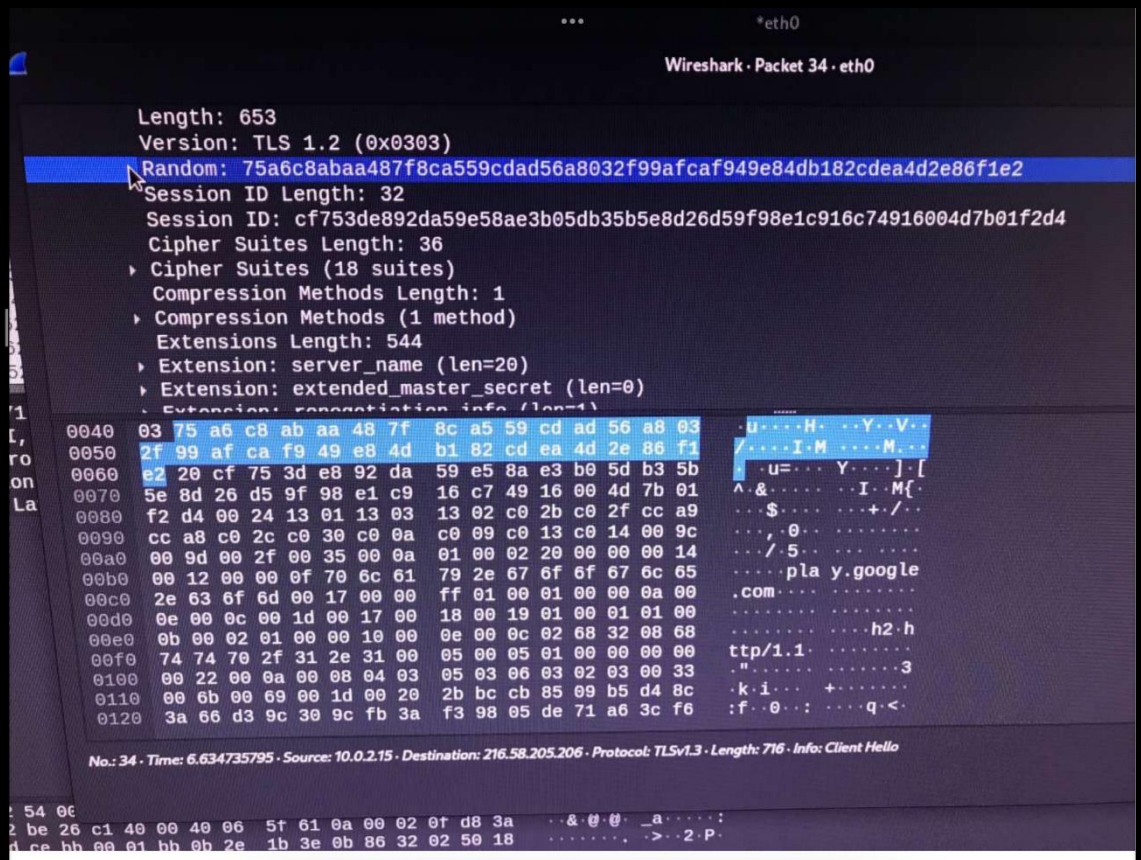
It is used to reduce volume



4. What is the session ID

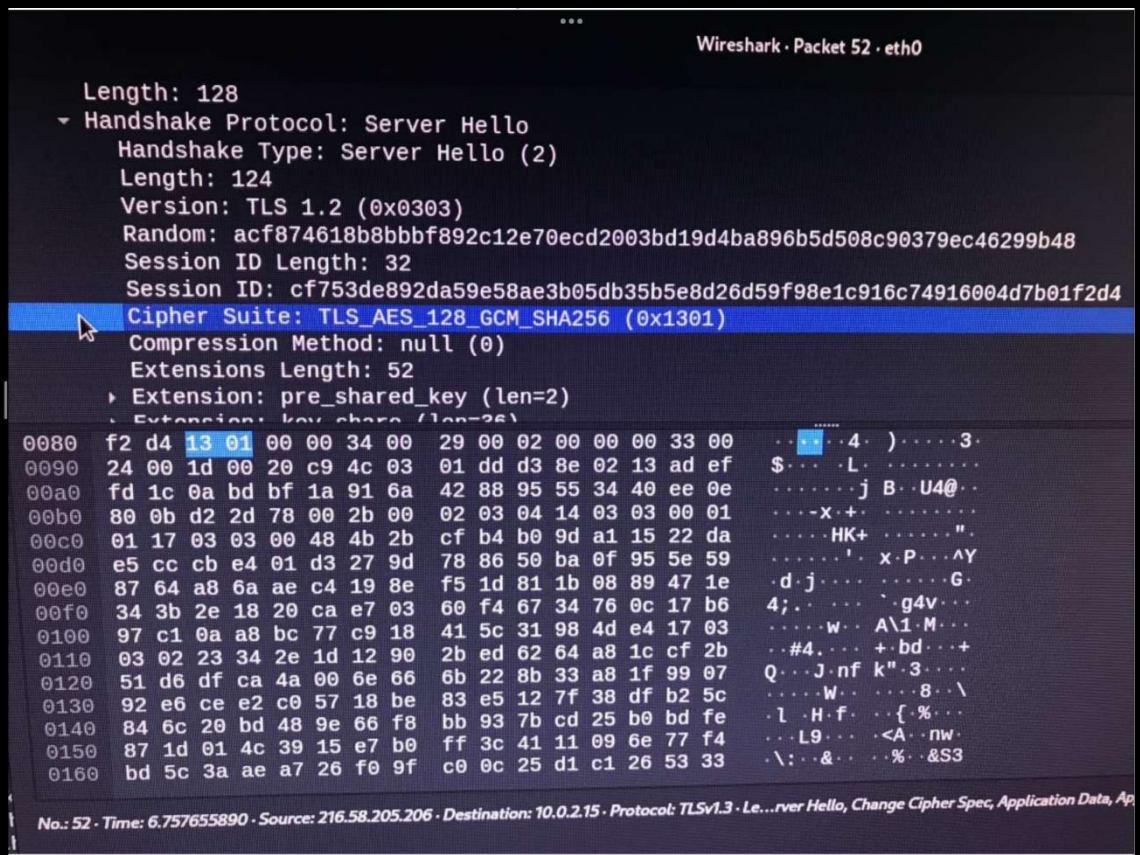


? 5. What is Random number



6. Inspect Server Hello, what did the server choose in cipher suite .

Choose → AES 128



	Source	Destination	Protocol	Length	Info
1	216.58.205.206	10.0.2.15	QUIC	240	Protected Payload (KP0), DCID=ed0a2f
14	10.0.2.15	216.58.205.206	QUIC	78	Protected Payload (KP0), DCID=eeb1885f1ad5620b
28	10.0.2.15	216.58.205.206	QUIC	73	Protected Payload (KP0), DCID=eeb1885f1ad5620b
90	216.58.205.206	10.0.2.15	TLSv1.3	915	Server Hello, Change Cipher Spec, Application Data, Application Data
96	10.0.2.15	216.58.205.206	TCP	54	47872 → 443 [ACK] Seq=839 Ack=862 Win=63714 Len=0
943	10.0.2.15	216.58.205.206	TLSv1.3	138	Application Data, Application Data
927	216.58.205.206	10.0.2.15	TCP	60	443 → 47872 [ACK] Seq=862 Ack=923 Win=65535 Len=0

bytes on wire (7320 bits), 915 bytes captured (7320 bits) on interface eth0, id 0

Ethernet II, Src: Intel E81C8 (08:00:27:00:08:00), Dst: Intel E81C8 (08:00:27:00:08:00)

Protocol: TCP

Source: 216.58.205.206, Destination: 10.0.2.15

Length: 915 bytes (7320 bits)

Marked: False

Ignored: False

in frame: eth:ethertype:ip:tcp:tls

Rule Name: TCP

Rule String: tcp

27 22 46 4f 52 54 00 12 35 02 00 00 45 00 ... FORT ... 5 ... E

39 b1 00 00 40 00 8b aa d9 3a cd ce 0a 00 ... 9 ... 0 ...

01 bb 00 00 0b 00 32 02 0b 2e 1a 84 50 18 ... 2 ... P

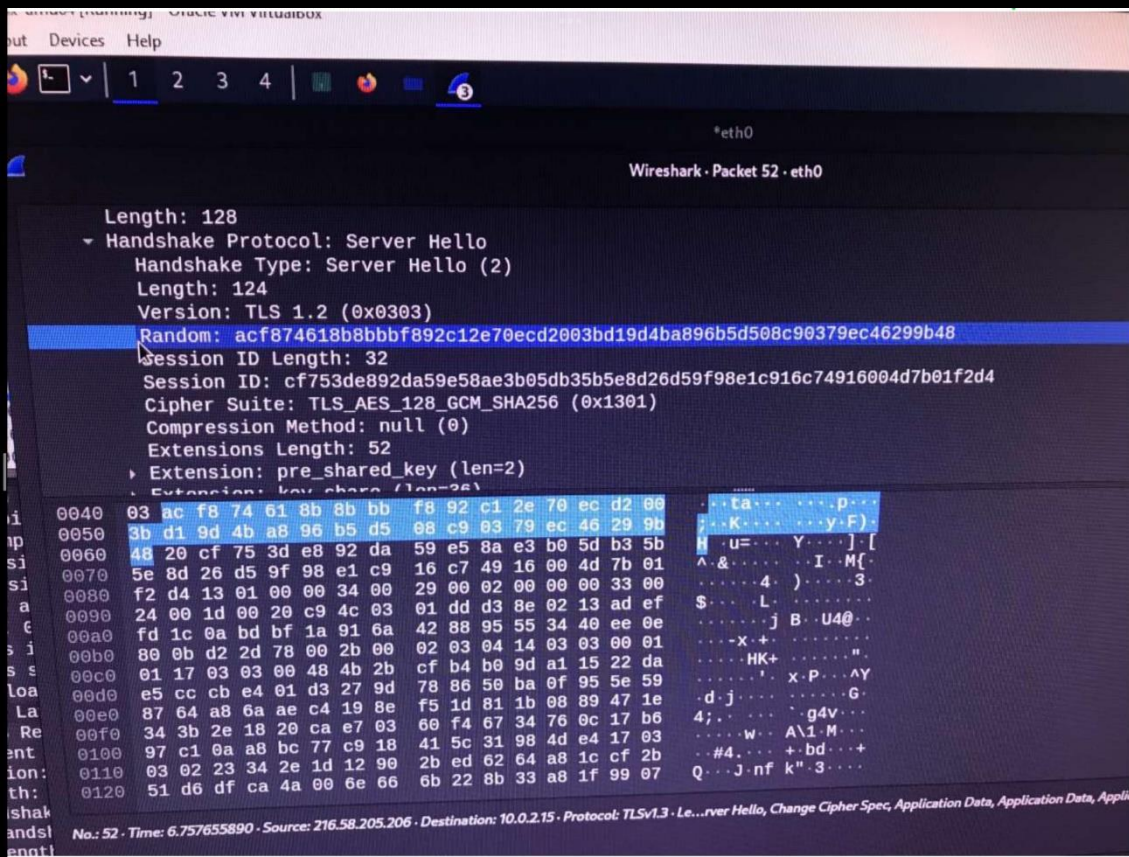
d9 44 00 00 16 03 03 00 00 02 00 00 7c 03 ... D ...

00 00 00 00 00 00 00 00 00 00 00 00 00 00 ... t ... p

7. Is the session ID the same from Server Hello? What does this refer to ??

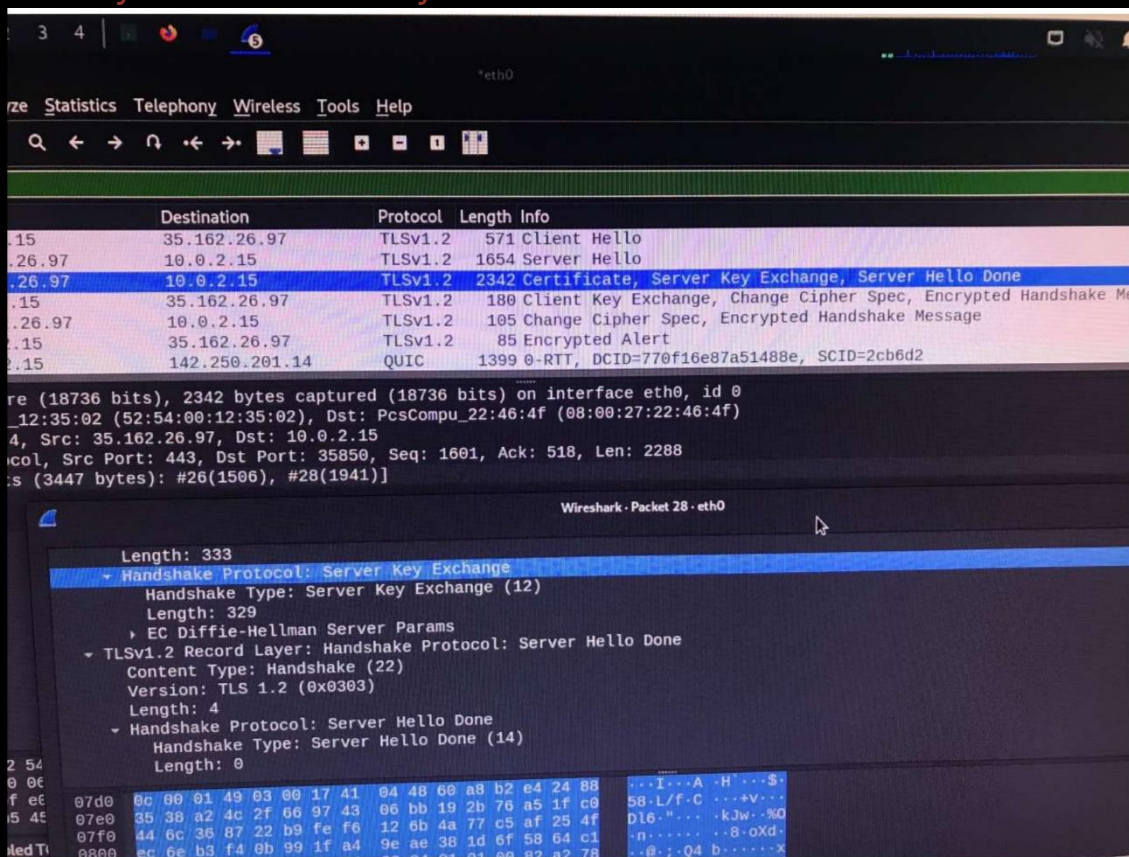
Yes... Because it is room for meet Server and Client

Refer to → unique number that a Web site's server assigns a specific user for the duration of that user's visit



9. Is there a certificate packet between Kku and your browser? Why?

No. Because Browser trusts the kku server because there was trust between them before. I've visited it before



10. Inspect Change cipher spec, do you see the contents of the message? If not, why?

No, because the msg is Encrypted

Time	Source	Destination	Protocol	Length	Info
23.6.452828162	10.0.2.15	35.162.26.97	TLSv1.2	571	Client Hello
26.7.065710652	35.162.26.97	10.0.2.15	TLSv1.2	1654	Server Hello
28.7.067231701	35.162.26.97	10.0.2.15	TLSv1.2	2342	Certificate, Server Key Exchange, Server Hello Done
30.7.085638529	10.0.2.15	35.162.26.97	TLSv1.2	188	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
32.7.087787497	35.162.26.97	10.0.2.15	TLSv1.2	185	Change Cipher Spec, Encrypted Handshake Message
34.7.103394662	10.0.2.15	35.162.26.97	TLSv1.2	85	Encrypted Alert
57.18.515498368	10.0.2.15	142.250.201.14	QUIC	1399	0-RTT, DCID=770f16e87a51488e, SCID=2cb6d2

TLSv1.2 Record Layer: Handshake Protocol: Client Key Exchange

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 70

Handshake Protocol: Client Key Exchange

Handshake Type: Client Key Exchange (16)

Length: 66

EC Diffie-Hellman Client Params

TLSv1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec

Content Type: Change Cipher Spec (20)

Version: TLS 1.2 (0x0303)

Length: 1

Change Cipher Spec Message

TLSv1.2 Record Layer: Handshake Protocol: Encrypted Handshake Message

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 40

Handshake Protocol: Encrypted Handshake Message

0070 0f 3d b5 36 22 10 8c 75 cd 5d 97 e6 01 52 88 7c

0080 f9 14 03 03 00 01 01 16 03 03 00 28 00 00 00 00

0090 00 00 00 00 e3 57 23 c8 21 5d 8e 13 ae 19 3c 87

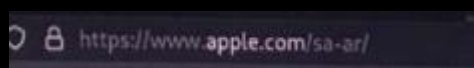
00a0 8e fc a9 b6 a3 3a 70 e0 f9 d9 d0 6b ae d5 01 2b

00b0 75 e4 76 58

11. Open any other website (https) and show the certificate of the website and explain .its contents captured by Wireshark .

Apple website<can I see the certificate . because I visit it first time , no trust between client and server before also they have public key , version ,Date The beginning and the end

And the CA it is use for Validate that original site (configuration)



Time	Source	Destination	Protocol	Length	Info
5149.96.319980864	10.0.2.15	23.44.140.22	TCP	54	60302 → 443 [ACK] Seq=518 Ack=5891 Win=59860 Len=0
5141.96.331047372	23.44.140.22	10.0.2.15	TLSv1.2	1654	Server Hello
5142.96.331097137	10.0.2.15	23.44.140.22	TCP	54	60406 → 443 [ACK] Seq=518 Ack=1601 Win=62780 Len=0
5143.96.331546679	23.44.140.22	10.0.2.15	TCP	2550	443 → 60406 [PSH, ACK] Seq=1601 Ack=518 Win=65535 Len=0
5144.96.331557283	10.0.2.15	23.44.140.22	TCP	54	60406 → 443 [ACK] Seq=518 Ack=4097 Win=61328 Len=0
5145.96.332468885	23.44.140.22	10.0.2.15	TLSv1.2	1654	Certificate [TCP segment of a reassembled PDU]
5146.96.332470510	10.0.2.15	23.44.140.22	TCP	54	60406 → 443 [ACK] Seq=518 Ack=5697 Win=59860 Len=0

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 5444

Handshake Protocol: Certificate

Handshake Type: Certificate (11)

Length: 5440

Certificates length: 5437

Certificates (5437 bytes)

Certificate length: 3836

Certificate: 30820bd630820ac0a00362016202144243534e0d3a4c2a7dca798b000000013ead9a7d30.. (id-at-commonName=itunes.apple.com, signedCertificate, algorithmIdentifier (sha256WithRSAEncryption)

3 . Building a simple Firewall on the client machine : (2 Marks)

- 1-Change the Default Chain Policies to Drop
- 2- Allow forwarding of TCP traffic on IP interface 10.0.1.2 (client) port 80 (HTTP) and port 443 (HTTPS) to go to 192.168.23.10
- 3- block a specific ip-address from reaching organization.
- 4- Incoming HTTP and HTTPS
- 5- Allow outbound DNS.

```
File Actions Edit View Help
└─$ sudo iptables -L
[sudo] password for kali:
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

(kali@kali)-[~]
└─$ sudo iptables -I INPUT -j DROP
(kali@kali)-[~]
└─$ sudo iptables -A INPUT -s 192.168.23.10 -j ACCEPT
(kali@kali)-[~]
└─$ sudo iptables -A FORWARD -p tcp --dport 80 -s 10.0.1.2/24 -d 192.168.23.10 -j ACCEPT
iptables: No chain/target/match by that name.
(kali@kali)-[~]
└─$ sudo iptables -A FORWARD -p tcp --dport 80 -s 10.0.1.2/24 -d 192.168.23.10 -j ACCEPT
(kali@kali)-[~]
└─$ sudo iptables -A FORWARD -p tcp --dport 443 -s 10.0.1.2/24 -d 192.168.23.10 -j ACCEPT
(kali@kali)-[~]
└─$ sudo iptables -A INPUT -s 161.255.130.130 -j DROP
(kali@kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
(kali@kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
(kali@kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 53 -j ACCEPT
(kali@kali)-[~]
└─$ Telnet 10.0.1.2
Telnet: command not found
(kali@kali)-[~]
└─$ telnet 10.0.1.2
Trying 10.0.1.2...
telnet: Unable to connect to remote host: Connection timed out
```

Instructions:

You must adhere to the instructions of the project:

1. Answer all the questions as required
2. The answer is by capturing the screen and attaching it to each question.
3. Each group consists of three students only.
4. Delay in project submission will be marked as zero.
5. Copying the answer between groups will be a case of cheating and the project mark will be deducted.